

```

# Install required libraries
!pip install opencv-python matplotlib scikit-image

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from skimage.filters import prewitt

# -----
# Upload Image
# -----
uploaded = files.upload()
image_path = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)

# -----
# SOBEL OPERATOR
# -----
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_mag = np.sqrt(sobelx**2 + sobely**2)
sobel_dir = np.arctan2(sobely, sobelx)

# -----
# PREWITT OPERATOR
# -----
prewitt_edges = prewitt(gray)

# -----
# SCHARR OPERATOR
# -----
scharrx = cv2.Scharr(gray, cv2.CV_64F, 1, 0)
scharry = cv2.Scharr(gray, cv2.CV_64F, 0, 1)
scharr_mag = np.sqrt(scharrx**2 + scharry**2)
scharr_dir = np.arctan2(scharry, scharrx)

# -----
# Sample Gradient Values (at center pixel)
# -----
h, w = gray.shape
cx, cy = h//2, w//2

print("----- SAMPLE GRADIENT VALUES (CENTER PIXEL) -----")
print("SOBEL -> Gx:", sobelx[cx,cy], "Gy:", sobely[cx,cy], "Magnitude:", sobel_mag[cx,cy])
print("SCHARR -> Gx:", scharrx[cx,cy], "Gy:", scharry[cx,cy], "Magnitude:", scharr_mag[cx,cy])

# -----
# Normalize for display
# -----
sobel_mag_disp = cv2.normalize(sobel_mag, None, 0, 255, cv2.NORM_MINMAX)
scharr_mag_disp = cv2.normalize(scharr_mag, None, 0, 255, cv2.NORM_MINMAX)

# -----
# Display Results
# -----
titles = [
    "Original Image", "Grayscale",
    "Sobel Magnitude", "Prewitt Edge", "Scharr Magnitude"
]

images = [
    img, gray,
    sobel_mag_disp, prewitt_edges, scharr_mag_disp
]

plt.figure(figsize=(12,6))
for i in range(len(images)):
    plt.subplot(2,3,i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')

plt.tight_layout()
plt.show()

```

Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-image in /usr/local/lib/python3.12/dist-packages (0.25.2)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

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OO.webp(image/webp) - 29682 bytes, last modified: 7/29/2026 - 100% done

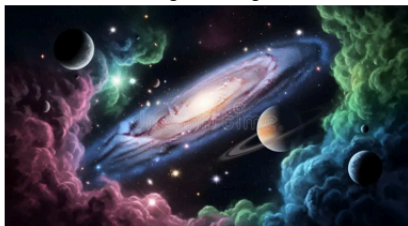
Saving OO.webp to OO.webp

----- SAMPLE GRADIENT VALUES (CENTER PIXEL) -----

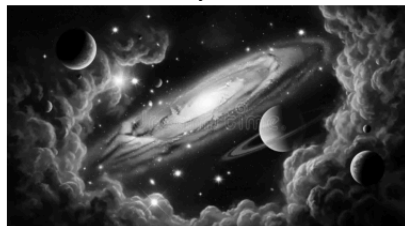
SOBEL -> Gx: 6.0 Gy: 10.0 Magnitude: 11.661903789690601

SCHARR -> Gx: 14.0 Gy: 82.0 Magnitude: 83.18653737234169

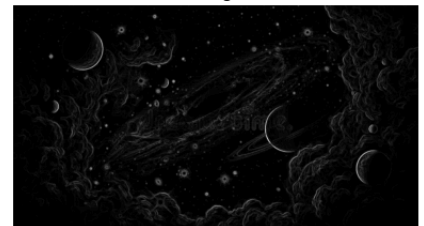
Original Image



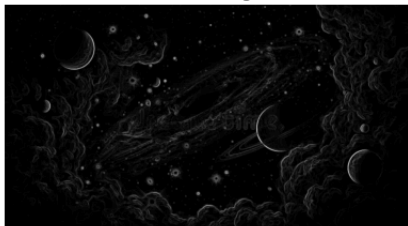
Grayscale



Sobel Magnitude



Prewitt Edge



Scharr Magnitude

